APPENDIX TO THE REQUIREMENT GATHERING DOCUMENT (RGD v2.0)

GAP-10

Appendix-J

Low Level Design (LLD)

School ERP

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1. Introduction

This document is Appendix-J to the Requirement Gathering Document (RGD v2.0) for the School ERP System, addressing Gap-10: Low Level Design (LLD). It converts the approved High Level Design (Appendix-I v1.0) into module-level technical design — internal components, processing flows, and cross-cutting mechanisms — for each of the 11 approved functional modules.

This document is implementation-neutral: it describes conceptual components (Controller, Service, Validator, Repository) and their responsibilities and interactions, without naming a specific programming language, framework, or library. Source code, SQL, stored procedures, API contracts, JSON payloads, and UI mockups are explicitly out of scope.

Per the governing low-credit rules for this task, this document does not re-analyze RGD v2.0 or regenerate any prior appendix. Every module, entity, Functional Requirement, Business Rule, and Non-Functional Requirement referenced below is reused by ID from Appendices E, F, G, H, and I, which remain the authoritative source.

1.1 Relationship to Appendix-I (HLD)

Appendix-I established the system's layered architecture (Presentation, Application/Business Logic, Data Access, Database), module boundaries, and cross-cutting architecture (Security, Logging, Notification, Reporting). This LLD does not redesign or reinterpret that baseline — it decomposes the Application/Business Logic layer of Appendix-I Section 7 into per-module internal components, and elaborates the conceptual flows (authentication, validation, error handling) that Appendix-I described at system level into module-applicable detail.

1.2 Intended Audience

- Development Team Leads — the primary audience, using this LLD as the direct input to component-level implementation planning.
- Solution Architects — validating that module-internal design remains consistent with the Appendix-H v1.1 data model and Appendix-I architecture.
- QA/Test Leads — using the Internal Processing Flow (Section 7) and Validation Flow (Section 8) as the basis for test-case design.

2. Scope

2.1 In Scope

- Module-internal component design (conceptual: Controller, Service, Validator, Repository, Event Publisher/Consumer, Background Job)
- Internal processing, validation, error-handling, and logging flows, described conceptually
- Cross-cutting technical mechanisms: session management, role-permission processing, notification processing, file handling, reporting, audit trail, background jobs, configuration
- Module dependency at the component level, extending Appendix-I Section 6's module-level view

2.2 Out of Scope

- Source code in any programming language
- SQL queries, CREATE TABLE statements, or stored procedures — Appendix-H v1.1 remains the authoritative (and also SQL-free) logical schema reference
- API specifications, endpoint definitions, or JSON request/response payloads
- UI mockups or screen-level design

This scope boundary is identical in spirit to the RGD-to-Appendix-E boundary and the Appendix-H-to-Physical-Design boundary already established earlier in this documentation set: this LLD describes internal design shape, not executable implementation.

3. Design Principles

- Separation of Concerns: each conceptual component (Section 5) has exactly one responsibility — a Controller never contains business logic, a Repository never contains validation logic.
- Single Source of Truth: reused unchanged from Appendix-I Section 4.3 — Student and Employee remain the sole authoritative identity records; no module-internal component maintains a duplicate copy.
- Business Rule Enforcement at the Service Layer: every one of the 71 Business Rules in Appendix-C v1.1 is enforced within a module's Service component, never left to the Controller or Presentation layer alone (consistent with Appendix-I Section 7.2 Layer Isolation Rationale).
- Dependency Direction: Controller → Service → Repository is a one-way dependency chain; a Repository never calls back into a Service, and a Service never calls a Controller.
- Event-Based Cross-Module Communication: where one module's action must affect another (e.g., Library fine posting into Fee Collection, Transport fee linkage), the interaction is modeled as an event publish/consume, not a direct cross-module Repository call — preserving the module boundary already established in Appendix-I Section 7.4.
- Configuration Over Hard-Coding: every value already flagged Client/Product Decision Required in Appendix-C v1.1 Section 3.5 and Appendix-F is read from the Configuration component (Section 19) at runtime, never embedded as a fixed value.
- Soft-Delete and Audit by Construction: every Repository component inherits the same soft-delete and audit-column behavior (Appendix-H v1.1 Sections 2.6-2.8), rather than each module re-implementing it independently.

4. Module-wise Internal Design

For each of the 11 approved modules: internal components, key processing steps, and implementation notes. Functional Requirement and Business Rule ranges are as catalogued in Appendix-E and Appendix-C v1.1 respectively; not restated field-by-field here.

4.1 Admission

Attribute	Detail
Functional Requirements	FR-01 to FR-05c
Business Rules	BR-ADM-001 to BR-ADM-008
Key Entities (Appendix-G/H)	Application, SeatAllocation (Appendix-G)

Internal Components

- AdmissionController — receives application submission/verification/confirmation requests
- AdmissionService — orchestrates the Application lifecycle (Submitted → Verified → Shortlisted/Waitlisted → Admitted)
- SeatAllocationService — enforces capacity/RTE-quota ceilings (BR-ADM-001, BR-ADM-003)
- DuplicateCheckValidator — enforces identity-uniqueness checks (BR-ADM-006)
- AdmissionRepository — persistence for Application and SeatAllocation

Key Processing Steps

- 1. Receive application data from Presentation layer
- 2. DuplicateCheckValidator checks Aadhaar/identity against existing Student records (BR-ADM-006)
- 3. SeatAllocationService checks class/RTE-quota capacity (BR-ADM-001, BR-ADM-003)
- 4. AdmissionService persists Application via AdmissionRepository
- 5. On confirmation, AdmissionService triggers Student Information module to create the Student record (FR-02)
- 6. NotificationService (Section 13) dispatches status update

Implementation Note: *The seat-hold-expiry / capacity race condition flagged in Appendix-H v1.1 Section 19 must be resolved here as an atomic operation spanning SeatAllocationService and AdmissionRepository — exact locking/transaction mechanism is Client/Product Decision Required.*

4.2 Student Information

Attribute	Detail
Functional Requirements	FR-06 to FR-09
Business Rules	BR-SIS-001 to BR-SIS-006
Key Entities (Appendix-G/H)	Student (Appendix-G ENT-SIS-001)

Internal Components

- StudentController — profile create/update/view requests
- StudentService — orchestrates profile completion, Draft→Active transition (BR-SIS-003)
- PromotionService — year-end promotion processing (BR-SIS-001)
- StudentValidator — mandatory-field and uniqueness validation (BR-SIS-002)
- StudentRepository — persistence for Student and PromotionRecord

Key Processing Steps

- 1. Receive profile data (from Admission handoff or Admin Staff edit)
- 2. StudentValidator checks mandatory fields, Aadhaar/admission-number uniqueness (BR-SIS-002, BR-SIS-003)
- 3. StudentService persists via StudentRepository; sets status Active only when complete
- 4. Downstream modules (Attendance, Fee, Examination, etc.) read the Student record via StudentService's query interface, never a duplicated copy

Implementation Note: *Student is the highest-fan-out module in the system (Appendix-D, Appendix-I Section 4.3/26.2) — any interface change to StudentService requires impact review across all 11 modules before implementation.*

4.3 Attendance

Attribute	Detail
Functional Requirements	FR-10 to FR-13
Business Rules	BR-ATT-001 to BR-ATT-007
Key Entities (Appendix-G/H)	AttendanceRecord, StaffAttendanceRecord (Appendix-G)

Internal Components

- AttendanceController — receives mark/correct requests
- AttendanceService — enforces single-state-per-period (BR-ATT-001) and lock/correction workflow (BR-ATT-002, BR-ATT-003)
- StaffAttendanceReconciliationJob — cross-checks staff absence against Leave (BR-ATT-005; see Section 17 Background Jobs)
- AttendancePercentageCalculator — computes rolling percentage feeding exam-eligibility flag (BR-ATT-006)
- AttendanceRepository — persistence

Key Processing Steps

- 1. Receive attendance-marking request for a class/period/date
- 2. AttendanceService checks for an existing conflicting record (BR-ATT-001)
- 3. If within edit window, direct save; if beyond window, route to ApprovalRequest (Appendix-H ENT-SYS-007) per BR-ATT-003
- 4. On Absent state, raise event consumed by NotificationService (Section 13)
- 5. AttendancePercentageCalculator recalculates on a scheduled basis (Section 17), not synchronously on every mark

Implementation Note: *Exact same-day edit window duration and minimum attendance percentage threshold remain Client/Product Decision Required, per Appendix-C v1.1 Section 3.5.*

4.4 Examination

Attribute	Detail
Functional Requirements	FR-17 to FR-21
Business Rules	BR-EXM-001 to BR-EXM-007
Key Entities (Appendix-G/H)	Exam, MarksRecord, ReportCard, PromotionRecord (Appendix-G)

Internal Components

- ExamController — schedule configuration, marks entry, report-card requests
- ExamConfigService — validates grading scheme against board affiliation (BR-EXM-007)
- MarksEntryService — range validation (BR-EXM-002) and anomaly flagging (BR-EXM-006)
- MarksLockService — enforces immutability after lock (BR-EXM-003) and re-evaluation workflow
- GradeCalculationEngine — computes grade/rank/GPA once all subjects locked (BR-EXM-001, BR-EXM-004)
- ReportCardService — generation and publish-gate enforcement (BR-EXM-001)
- ExamRepository — persistence

Key Processing Steps

- 1. Academic Head configures Exam via ExamConfigService
- 2. Teacher submits marks; MarksEntryService validates range and anomaly
- 3. On completion of all subjects, GradeCalculationEngine triggers automatically (event-driven, not polled)
- 4. ReportCardService generates and gates publication on the lock-completeness check (BR-EXM-001)
- 5. Publish event triggers NotificationService (Section 13)

Implementation Note: *GradeCalculationEngine's trigger condition ("all subjects locked") must be evaluated transactionally to avoid a race between the last subject lock and calculation trigger — implementation approach Client/Product Decision Required.*

4.5 Fees

Attribute	Detail
Functional Requirements	FR-22 to FR-26b
Business Rules	BR-FEE-001 to BR-FEE-008
Key Entities (Appendix-G/H)	FeeHead, FeeStructure, Invoice, Payment, ScholarshipWaiver (Appendix-G)

Internal Components

- FeeConfigController/Service — fee head and structure configuration
- InvoiceGenerationService — scheduled and ad-hoc invoice creation, waiver application (BR-FEE-005)
- PaymentService — records online/offline payment, enforces transaction-reference uniqueness (BR-FEE-006)
- InvoiceLockValidator — enforces post-payment immutability (BR-FEE-001)
- RefundService — Finance-Team-exclusive refund/void processing (BR-FEE-002)
- DefaulterEscalationJob — scheduled overdue detection (BR-FEE-008; see Section 17)
- FeeRepository — persistence

Key Processing Steps

- 1. InvoiceGenerationService creates invoices per schedule, applying FeeStructure and any ScholarshipWaiver
- 2. Parent-initiated payment routes through PaymentService, which validates the gateway transaction reference for uniqueness before committing (BR-FEE-006)
- 3. On payment success, InvoiceLockValidator marks the invoice immutable (BR-FEE-001)
- 4. Corrections after locking route through a CreditNoteService, never a direct Invoice edit
- 5. RefundService checks the requesting user's Role before processing (BR-FEE-002)

Implementation Note: *This module carries the highest financial risk in the system (Appendix-I Section 5.1.6); the BR-ADM-001/BR-ADM-007-style atomic-transaction discipline noted for Admission applies equally here to PaymentService's duplicate-prevention check.*

4.6 Library

Attribute	Detail
Functional Requirements	FR-27 to FR-29
Business Rules	BR-LIB-001 to BR-LIB-006
Key Entities (Appendix-G/H)	Book, BookIssue (Appendix-G)

Internal Components

- CatalogController/Service — Book create/search (FR-27, FR-29)
- IssueReturnService — eligibility checks (BR-LIB-001, BR-LIB-004, BR-LIB-005) and fine calculation (BR-LIB-002)
- ReservationQueueService — FIFO reservation handling (BR-LIB-006)
- LibraryRepository — persistence

Key Processing Steps

- 1. Borrower requests a book; IssueReturnService checks max-books limit, Reference-classification restriction, and outstanding-fine threshold before allowing issue
- 2. On return, IssueReturnService calculates any overdue fine and posts it to Fee Collection via a cross-module event, not a direct database write into Fee's tables
- 3. If a reservation queue exists for the returned title, ReservationQueueService notifies the next borrower in FIFO order

Implementation Note: *Fine-per-day rate and max-books-per-student limit remain Client/Product Decision Required.*

4.7 Transport

Attribute	Detail
Functional Requirements	FR-30 to FR-32
Business Rules	BR-TRN-001 to BR-TRN-006
Key Entities (Appendix-G/H)	Route, Vehicle, TransportAllocation (Appendix-G)

Internal Components

- RouteController/Service — route/vehicle configuration
- AllocationService — capacity and single-active-allocation enforcement (BR-TRN-001, BR-TRN-002)
- EmergencyContactValidator — mandatory-field check before allocation (BR-TRN-004)
- LiveTrackingService — GPS feed ingestion and staleness detection (BR-TRN-003)
- TransportFeeLinkService — cross-module event publisher to Fee Collection (FR-32)
- TransportRepository — persistence

Key Processing Steps

- 1. Parent requests allocation; AllocationService checks route capacity and existing active allocation
- 2. EmergencyContactValidator blocks allocation if no contact is on file
- 3. On confirmed allocation, TransportFeeLinkService raises an event consumed by Fee Collection's InvoiceGenerationService
- 4. LiveTrackingService ingests GPS device data at the configured interval and flags stale data if the feed stops (BR-TRN-003)

Implementation Note: *GPS refresh interval and driver/vehicle licensing data source remain Client/Product Decision Required.*

4.8 HR & Payroll

Attribute	Detail
Functional Requirements	FR-33 to FR-36
Business Rules	BR-HR-001 to BR-HR-007
Key Entities (Appendix-G/H)	Employee, Department, Designation, PayrollRun, LeaveRequest (Appendix-G)

Internal Components

- EmployeeController/Service — staff master maintenance, exit processing (BR-HR-006)
- AccessRevocationService — triggers User deactivation on exit (BR-HR-002)
- LeaveService — balance validation (BR-HR-004) and approval routing
- PayrollService — gates on Attendance closure (BR-HR-001), prevents duplicate runs (BR-HR-003)
- StatutoryDeductionCalculator — PF/ESI/PT/TDS application (BR-HR-005)
- PayslipService — immutable payslip issuance (BR-HR-007)
- HRRepository — persistence

Key Processing Steps

- 1. HR confirms staff exit; AccessRevocationService immediately deactivates the linked User record (BR-HR-002) — this call is synchronous and blocking, not eventually-consistent, given its security criticality
- 2. PayrollService checks Attendance module's month-close status before allowing a run (BR-HR-001)
- 3. StatutoryDeductionCalculator applies current slabs; PayslipService issues and locks the payslip (BR-HR-007)
- 4. Post-issuance corrections route through a supplementary adjustment record, never a direct payslip edit

Implementation Note: *Current PF/ESI/Professional Tax statutory slabs remain Client/Product Decision Required and must be sourced from an externally maintained, versioned configuration (Section 19), not hard-coded.*

4.9 Communication

Attribute	Detail
Functional Requirements	FR-37 to FR-39
Business Rules	BR-COM-001 to BR-COM-005
Key Entities (Appendix-G/H)	Circular, NotificationLog (Appendix-G)

Internal Components

- NotificationController/Service — the shared dispatch entry point consumed by every other module (Section 13)
- ChannelAdapter (SMS/Email/Push) — one adapter per external gateway (Appendix-I Section 11)
- BulkAuthorizationValidator — threshold-based authorization gate (BR-COM-002)
- RelationshipValidator — parent-teacher messaging scope check (BR-COM-001)
- CommunicationRepository — persistence for Circular and NotificationLog

Key Processing Steps

- 1. Any module raises a notification event with recipient, channel preference, and content
- 2. NotificationService checks BulkAuthorizationValidator if the audience exceeds the configured threshold (BR-COM-002)
- 3. ChannelAdapter dispatches via the appropriate external gateway; failures are logged to NotificationLog, not silently retried indefinitely
- 4. Emergency Alert message type bypasses the standard queue entirely (BR-COM-003)

Implementation Note: *This is the module most exposed to the cross-module single-point-of-failure risk already documented in Appendix-D and Appendix-I Section 26.2 — its internal design must isolate ChannelAdapter failures per-channel so one gateway's outage does not block another's dispatch.*

4.10 Reports

Attribute	Detail
Functional Requirements	FR-40 to FR-42
Business Rules	BR-RPT-001 to BR-RPT-005
Key Entities (Appendix-G/H)	None of its own beyond dashboard configuration (Appendix-I Section 5.1.11)

Internal Components

- DashboardController/Service — role-scoped widget aggregation (BR-RPT-002 read-only enforcement)
- ReportBuilderService — custom field/filter selection with field-level authorization (BR-RPT-001)
- ProvisionalDataLabeller — flags reports drawing on unlocked source data (BR-RPT-004)
- ExportService — background-job handling for large exports (Appendix-F NFR-RPTP-001)
- TrendAnalyticsService — historical aggregation over closed AcademicSession data

Key Processing Steps

- 1. Requesting user's Role determines available dashboard widgets and report-builder fields
- 2. Aggregation reads directly from source modules' repositories (read-only) — Reports never writes back to source data (BR-RPT-002)
- 3. ProvisionalDataLabeller checks each contributing data source's lock status before rendering
- 4. Exports beyond the row-count threshold are queued as background jobs (Section 17) rather than blocking the request

Implementation Note: *A pure consumption layer by design (Appendix-I Section 14) — no new business logic is introduced here beyond aggregation, filtering, and access-scoping.*

4.11 Administration

Attribute	Detail
Functional Requirements	N/A (cross-cutting system entities)
Business Rules	BR-HR-002 (access revocation, shared); no module-specific rules
Key Entities (Appendix-G/H)	User, Role, Guardian, AuditLog, Configuration, Document, ApprovalRequest (Appendix-G ENT-SYS series)

Internal Components

- UserController/Service — account provisioning, deactivation, password reset
- RoleService — Role/Permission management via the RolePermission junction (Appendix-H v1.1 Section 6)
- ConfigurationService — runtime key-value configuration read/write (Section 19)
- AuditLogService — the single, centralized audit-write path consumed by every other module's Service layer
- ApprovalRequestService — generic approval/override workflow engine (Appendix-H ENT-SYS-007)
- DocumentService — polymorphic file storage/reference management (Section 14)

Key Processing Steps

- 1. Every other module's Service layer calls AuditLogService directly after any Create/Update/Delete/Approve action — this is not an optional or best-effort call, but a required step in each such transaction
- 2. ConfigurationService is read by every module at startup/request-time for policy values (rates, thresholds, SLAs) rather than those values being hard-coded per module
- 3. ApprovalRequestService is invoked wherever a Business Rule requires elevated authorization (BR-ATT-003, BR-EXM-003, BR-HR-004, BR-SIS-004, BR-FEE-002)

Implementation Note: *This is the only module with cross-cutting, system-wide reach (Appendix-I Section 8.1); its components are conceptually shared libraries/services consumed by all other modules' internal designs, not a peer module in the same sense as Admission or Fees.*

5. Component Breakdown

Every module in Section 4 is composed from the same six conceptual component types, applied consistently rather than reinvented per module.

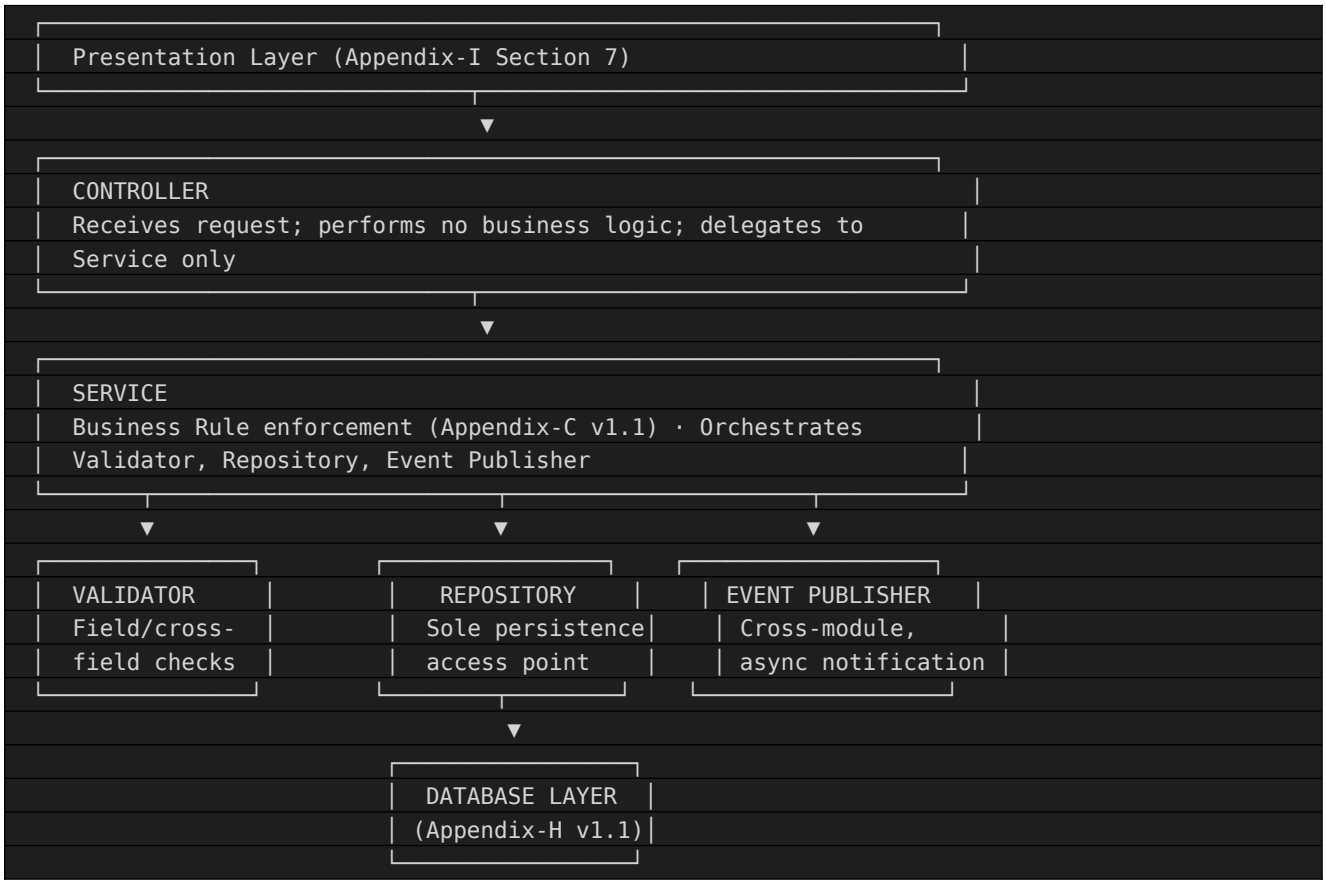
Component Type	Responsibility
Controller	Receives requests from the Presentation layer boundary (Appendix-I Section 7.3); performs no business logic itself, only delegates to Service
Service	Owns business logic and Business Rule enforcement (Appendix-C v1.1); orchestrates calls to Validator, Repository, and cross-module event publishing
Validator	Stateless component performing field-level and cross-field validation (Appendix-E Field 16, per FR) before a Service commits a change
Repository	Sole component permitted to perform persistence operations; enforces the entity's Primary/Foreign Key and soft-delete rules (Appendix-H v1.1 Sections 2.3-2.8)
Event Publisher / Consumer	Used for cross-module, asynchronous notifications (e.g., Fee module publishing a defaulter event consumed by Communication) — decouples modules per Appendix-I Section 7.4
Background Job	Scheduled or triggered process not directly invoked by a user request (Section 17)

5.1 Component Reuse Across Modules

Some components are module-specific (e.g., MarksLockService exists only in Examination); others are shared/system-wide, provided by the Administration module (Section 4.11) and consumed by every other module: AuditLogService, NotificationService, ConfigurationService, ApprovalRequestService, and DocumentService. This mirrors the shared-service architectural principle already established in Appendix-I Section 4.3.

6. Controller-Service-Repository Interaction (Conceptual Only)

Conceptual layer interaction pattern applied uniformly across all 11 modules' internal designs (Section 4).



A Controller never calls a Repository directly, and a Validator never calls a Repository at all — this ordering ensures Business Rules are always evaluated before a persistence attempt, not after.

7. Internal Processing Flow

Generic request-to-response flow, applicable to any module's primary write operation (e.g., Admission's Application submission, Fee's Payment recording, Examination's Marks entry).

```
Request received by Controller
        |
        ▼
Controller delegates to Service (no logic in Controller itself)
        |
        ▼
Service invokes Validator (Section 8)
        |
    ┌───┴─────┐
    ▼         ▼
 Valid      Invalid ──► Return validation error (Section 9) to Controller
    |
    ▼
Service evaluates applicable Business Rule(s) (Appendix-C v1.1)
        |
    ┌───┴─────┐
    ▼         ▼
 Allowed    Blocked ──► Return business-rule error (Section 9) to Controller
    |
    ▼
Service calls Repository to persist (Section 10 Logging Flow runs alongside)
        |
        ▼
Repository writes with audit columns populated (Appendix-H v1.1 Section 2.6)
        |
        ▼
AuditLogService called (Section 16) — mandatory, not optional
        |
        ▼
If cross-module effect exists, Event Publisher raises event (Section 13/14 consumers)
        |
        ▼
Success response returned to Controller → Presentation
```

8. Validation Flow

Two-tier validation, consistent with Appendix-E Field 36 (Validation Messages) and Field 16 (Validation Rules) per Functional Requirement.

8.1 Client-Side (Presentation) Validation

Advisory only — improves user experience by catching obvious errors (empty mandatory fields, malformed email) before a request is even sent. Never trusted as the authoritative check, consistent with Appendix-I Section 7.3's Presentation-to-Application boundary principle.

8.2 Service-Layer (Authoritative) Validation

- Mandatory Field Validation: every attribute marked Mandatory=Y in Appendix-G's Attribute Catalogue is checked.
- Format Validation: pattern/type checks (e.g., 10-digit mobile number, Aadhaar checksum) per Appendix-G Field "Validation Rule".
- Range Validation: numeric/date bounds (e.g., MarksRecord.marks_obtained ≤ max_marks).
- Uniqueness Validation: checked against the entity's Unique Constraints (Appendix-H v1.1 Section 4, Field 10) before Repository commit.
- Cross-Field Validation: e.g., LeaveRequest.end_date ≥ start_date.
- Business Rule Validation: the 71 rules in Appendix-C v1.1, evaluated by the Service layer after basic field validation passes, per the Internal Processing Flow (Section 7).

8.3 Validation Failure Handling

A validation failure short-circuits the Internal Processing Flow (Section 7) immediately — no partial persistence occurs. The specific failure message returned reuses the Validation Messages already defined per requirement in Appendix-E Field 36, not newly invented per module.

9. Error Handling Strategy

Reuses the standardized pattern already defined once in Appendix-E (Field 37, applied consistently to all 47 requirements) and reaffirmed as Appendix-F NFR-ERR-001.

Error Category	Handling Approach
Validation Error	Inline, field-level error message returned to Controller/Presentation; no persistence attempted
Business Rule Violation	Specific error identifying the violated rule (Appendix-C v1.1 ID); logged if the attempt itself is security-relevant (e.g., repeated unauthorized refund attempts)
Integration/External System Failure	Caught at the Event Publisher/ChannelAdapter boundary (Section 6); user-facing operation is not blocked, but the failure is logged (Section 10) and, where applicable, queued for retry
System/Unexpected Error	Generic, non-revealing message to the user; full technical detail logged for IT Admin; no partial/corrupt data committed (Appendix-H v1.1 Section 2.9's transactional consistency principle applies)
Concurrency Conflict	Detected via the version field (Appendix-H v1.1 Section 2.8); the conflicting user is notified and asked to refresh, rather than one write silently overwriting another

10. Logging Flow

Every Service-layer component logs through the same AuditLogService (Administration module, Section 4.11), consistent with Appendix-I Section 15 and Appendix-F's Logging NFR series.

```
Service completes a Create/Update/Delete/Approve action
        |
        ▼
Service calls AuditLogService.record(entity_name, record_id, action,
                                     performed_by, old_value, new_value, reason)
        |
        ▼
AuditLogService writes an immutable AuditLog record (Appendix-H ENT-SYS-004)
        |
        ▼
Available for query via the Consolidated Audit Trail interface
(Appendix-F NFR-LOG-010), filtered by entity_name for module-specific views
```

This call is not optional or best-effort at the Service layer — a Service implementation that completes a state-changing action without the corresponding AuditLogService call is considered incomplete against this design.

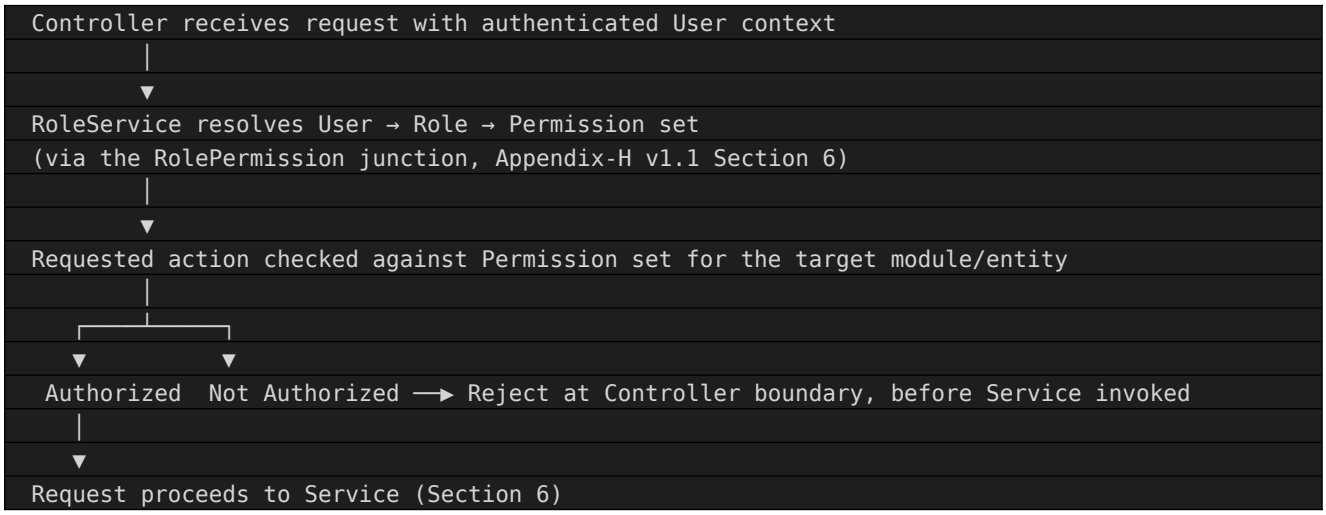
11. Session Management

Reuses Appendix-I Section 9 (Authentication & Authorization Flow) and Appendix-F NFR-SESS-001 at the component level.

- A session/token is issued by the Administration module's UserService upon successful authentication and is validated by the Controller layer (Section 6) on every subsequent request before delegating to Service.
- Session/token expiry after the configured idle period (NFR-SESS-001, Client/Product Decision Required for exact duration) is enforced at the Controller boundary, not left to individual module Services to check independently — this avoids inconsistent session handling across the 11 modules.
- On password reset or access revocation (BR-HR-002), all active sessions for the affected User are invalidated by UserService, consistent with Appendix-I Section 9.2.

12. Role Permission Processing

Reuses Appendix-E Field 38 (Role Permissions per requirement) and Appendix-I Sections 8-9 at the component level.



This check occurs at the Controller boundary specifically so that an unauthorized request never reaches Service-layer business logic at all — consistent with Appendix-I Section 7.3's Presentation-to-Application boundary being the single enforcement point for authorization.

13. Notification Processing

Reuses the Communication module's internal design (Section 4.9) and Appendix-I Section 13 at the component level.

- Any module's Service component raises a notification event (Section 6 Event Publisher) rather than calling an external gateway directly — this indirection is what allows Communication's ChannelAdapter components to be swapped per vendor (Appendix-I Section 11.1) without touching the 10 other modules' internal code.
- NotificationService (Section 4.9) resolves the event into a channel-specific dispatch (SMS/Email/Push) and logs the outcome to NotificationLog regardless of success or failure.
- Emergency Alert events bypass the standard event queue entirely (BR-COM-003), routing directly to all configured ChannelAdapters in parallel.

14. File Handling Process

Reuses the Document entity design (Appendix-G/H ENT-SYS-006) and Appendix-I Section 12 at the component level.

```
Module Service receives a file (upload) or generates one (e.g., report card)
        |
        ▼
DocumentService (Administration module) validates type/size and scans for
malicious content (Appendix-F NFR-FUS-001) before accepting
        |
        ▼
File persisted to File/Object Storage Tier (Appendix-I Section 17);
DocumentService records a Document entity with polymorphic owner_type/owner_ref_id
        |
        ▼
Retrieval always passes through DocumentService's authorization check
(Appendix-I Section 12.3) — never a direct storage-path reference exposed to the client
```

15. Reporting Process

Reuses the Reports module's internal design (Section 4.10) and Appendix-I Section 14 at the component level.

- DashboardService aggregates read-only data directly from each source module's Repository query interface — it never writes back, consistent with BR-RPT-002.
- ReportBuilderService filters available fields by the requesting User's Role before rendering the selection UI-facing data (field-level authorization, BR-RPT-001), not merely after a field has already been selected.
- ExportService checks the row-count threshold (Appendix-F NFR-RPTP-001); requests below threshold are handled synchronously, requests above are handed to a Background Job (Section 17) with a completion notification via NotificationService (Section 13).

16. Audit Trail Processing

Elaborates Section 10 (Logging Flow) specifically for the audit-trail use case, reusing Appendix-I Section 15 and Appendix-F NFR-AUDIT-001.

- AuditLogService (Administration module) is the single write path for all audit records across all 11 modules — no module implements its own parallel audit mechanism.
- Audit records are immutable once written; AuditLogService exposes no update or delete operation, consistent with Appendix-H v1.1 Section 15 (write-once, read-many).
- Module-specific audit views (e.g., "Fee Transaction Logging", Appendix-F NFR-LOG-002) are implemented as filtered queries against the single AuditLog store (filtered by entity_name), not as separate physical logs per module.

17. Background Jobs

Scheduled or event-triggered processes that run independently of a direct user request, reused from the relevant module sections above and Appendix-F's NFR targets.

Job	Module	Trigger	Purpose
DefaulterEscalationJob	Fees	Daily schedule	Overdue detection and reminder escalation (BR-FEE-008)
StaffAttendanceReconciliationJob	Attendance	Daily schedule (end-of-day)	Cross-check staff absence against Leave (BR-ATT-005)
AttendancePercentageRecalcJob	Attendance	Nightly schedule	Recompute rolling attendance percentage (feeds BR-ATT-006)
BackupJob	Administration	Daily full + continuous log	Per Appendix-F NFR-BKP-001
ArchivalJob	Administration (cross-module)	Post academic-session close	Move closed-session data to archival tier per Appendix-H v1.1 Section 15
NotificationRetryJob	Communication	On dispatch failure	Retry per configured policy; does not retry indefinitely
LargeExportJob	Reports	On-demand (threshold-triggered)	Background processing for exports >10,000 rows (NFR-RPTP-001)
BackupIntegrityCheckJob	Administration	Monthly schedule	Restore-test verification per NFR-BKP-005

18. Module Dependency

Component-level view of the module dependency chain already established in Appendix-D and reaffirmed in Appendix-I Section 6.

```
Admission.AdmissionService
        │ (on confirmation)
        ▼
StudentInfo.StudentService  ◄── consumed by nearly every other module's Service layer
        │
        ├──► Attendance.AttendanceService
        ├──► Timetable (Appendix-I; not separately detailed here — no distinct LLD module req
uested)
        ├──► Fees.InvoiceGenerationService
        ├──► Library.IssueReturnService
        └──► Transport.AllocationService

Attendance.AttendanceService ──► Examination.MarksEntryService (eligibility check)
Attendance.StaffAttendanceReconciliationJob ──► HRPayroll.PayrollService (gating)
Library.IssueReturnService ──► Fees.InvoiceGenerationService (fine posting, event-based)
Transport.TransportFeeLinkService ──► Fees.InvoiceGenerationService (event-based)
Examination / Fees / HRPayroll ──► Reports.DashboardService (read-only aggregation)
Attendance / Fees / Examination ──► Communication.NotificationService (event-based)
```

Note: Timetable & Scheduling was not listed among the 11 modules requested for Section 4 detail in this LLD's governing instruction (it is folded into the Attendance and Examination flows above as a dependency, consistent with its role in Appendix-I); its own internal component design, if required, would follow the identical pattern as the 11 modules in Section 4.

19. Configuration Components

Reuses the Configuration entity (Appendix-G/H ENT-SYS-005) and Appendix-F NFR-CFG-001 at the component level.

- ConfigurationService provides a single read (and IT-Admin-only write) interface over the Configuration key-value store, consumed by every module's Service layer for policy values.
- Every value catalogued as Client/Product Decision Required in Appendix-C v1.1 Section 3.5 (late-fee rate, attendance threshold, GPS interval, session timeout, etc.) is modeled as a Configuration entry, not a fixed value in any Service's internal logic.
- Configuration changes are themselves audited (Section 16), consistent with Appendix-F NFR-LOG-009 (Configuration Change Logging).

20. Exception Handling

Technical exception categories, distinct from (but consistent with) the business-facing Error Handling Strategy in Section 9.

- Validation Exception: raised by a Validator component (Section 8); always caught at the Service layer and translated into the Section 9 Validation Error response.
- Business Rule Exception: raised by a Service component when a Business Rule (Appendix-C v1.1) blocks an action; carries the specific rule ID for traceability.
- Integration Exception: raised by an Event Publisher/ChannelAdapter (Section 6/13) on external-system failure; never allowed to propagate uncaught into the Controller layer, per Appendix-I Section 11.2's independent-failure-isolation principle.
- Concurrency Exception: raised by a Repository component on a version-field mismatch (Appendix-H v1.1 Section 2.8); always caught at the Service layer and translated into the Section 9 Concurrency Conflict response.
- Unhandled/System Exception: any exception not caught by the above is logged in full (Section 10) and translated into the generic Section 9 System Error response before reaching the Controller — no internal exception detail (stack trace, internal identifiers) is ever returned to the Presentation layer.

21. Performance Considerations

Component-level implications of the targets already set in Appendix-F and reaffirmed in Appendix-I Section 20.

- Repository components for high-volume entities (AttendanceRecord, MarksRecord, Invoice/Payment, NotificationLog, AuditLog) must use the indexing already specified in Appendix-H v1.1 Section 12; a Service must not perform full-table scans against these Repositories under any code path.
- GradeCalculationEngine (Examination) and StatutoryDeductionCalculator (HR & Payroll) are computation-heavy Service components; both should be designed to run once per triggering event, not repeatedly recomputed on every read (results are persisted, then read).
- ExportService and LargeExportJob (Sections 15, 17) exist specifically so that a single expensive report request cannot degrade Page Load Time (NFR-PERF-001) for concurrent users on unrelated modules.
- Caching (Appendix-I Section 17, NFR-CACHE-001) is applied at the Service layer for slow-changing master data (Class, Section, Subject, published Timetable) — Repository components for these entities should support a cache-aware read path.

22. Security Considerations

Component-level implications of Appendix-F's Security NFR series and Appendix-I Section 16.

- Every Controller enforces the Role Permission check (Section 12) before delegating to Service — this is non-negotiable and identical across all 11 modules' Controllers.
- Sensitive/PII fields (Appendix-G/H Sensitive Data Matrix — e.g., Student.medical_info, Employee.salary_structure_json, User.password_hash) require Repository-level field encryption support (NFR-ENC-001); a Service must never log these values in plaintext (Section 10 AuditLogService must redact or reference rather than store raw Sensitive values where the field itself is the sensitive data).
- File upload handling (Section 14) validates type/size and scans for malicious content before DocumentService accepts a file, per NFR-FUS-001.
- CSRF/XSS/SQL Injection protections (NFR-SEC-001/002/003) apply at the Controller and Repository boundaries respectively — Controllers reject unvalidated state-changing requests without a valid anti-CSRF token; Repositories never construct dynamic queries from unsanitized input.
- Polymorphic association integrity (owner_type/owner_ref_id pattern used by User, Document, ApprovalRequest, NotificationLog, BookIssue) must be validated at the Service layer, since it cannot be enforced by a Repository-level foreign-key constraint alone — this is the same risk already flagged in Appendix-H v1.1 Section 19, restated here as a Service-layer implementation obligation.

23. Maintainability Guidelines

Reuses Appendix-F NFR-MAINT-001, NFR-EXT-001, and NFR-CFG-001 at the component level.

- A module's Service components should depend on other modules' Service interfaces (or published events), never on another module's Repository or Database entities directly — this is what makes the monolith-vs-services physical-implementation choice (Appendix-I Section 4.2) still open at this design stage without rework.
- New modules (e.g., a future Hostel or Alumni module, per Appendix-H v1.1 Section 18) should be addable by implementing the same six component types (Section 5) and consuming the shared Administration-module services (AuditLogService, NotificationService, ConfigurationService, DocumentService, ApprovalRequestService), without modifying existing modules' internals.
- Configuration-driven behavior (Section 19) is preferred over conditional code branches wherever a value is already flagged Client/Product Decision Required, so that a future policy change (e.g., a new late-fee formula) does not require a code deployment.

24. Traceability to Appendix-I

LLD Section	Appendix-I (HLD) Source
4. Module-wise Internal Design	Appendix-I Section 5 (Major Modules) and Section 5.1 (Module Descriptions)
6. Controller-Service-Repository Interaction	Appendix-I Section 7 (Layered Architecture) and Section 7.3-7.4
9. Error Handling Strategy	Appendix-I Section 7.1 (Layer Responsibilities) — Error Handling row
10-11-16. Logging, Session, Audit	Appendix-I Sections 9, 15
12. Role Permission Processing	Appendix-I Sections 8-9
13. Notification Processing	Appendix-I Section 13
14. File Handling Process	Appendix-I Section 12
15. Reporting Process	Appendix-I Section 14
17. Background Jobs	Appendix-I Sections 13, 15, 17-20 (implied by NFR targets)
18. Module Dependency	Appendix-I Section 6
19. Configuration Components	Appendix-I Section 21 (Institute Isolation) and general Configuration principle (Appendix-I Section 4.3)
21-22. Performance, Security	Appendix-I Sections 16, 18, 20

Every LLD section above elaborates, and does not contradict, its corresponding Appendix-I section. No new architectural decision is introduced at this stage — only internal decomposition of decisions already made.

25. Validation Checklist

Check	Result
All 27 required sections present	PASS — Sections 1 through 27 as specified
All 11 requested modules detailed in Section 4	PASS — Admission, Student Information, Attendance, Examination, Fees, Library, Transport, HR & Payroll, Communication, Reports, Administration
No source code, SQL, or CREATE TABLE included	PASS
No API specifications or JSON payloads included	PASS
No UI mockups included	PASS
No sequence or class diagrams (UML) included	PASS — only Component, Module Interaction, Internal Processing Flow, and Layer Interaction diagrams, all ASCII
No new modules, entities, business rules, roles, or integrations invented	PASS — every reference traces to Appendix-E, F, G, H, or I by ID
Every unknown implementation decision flagged	PASS — marked Client/Product Decision Required consistently with Appendix-C v1.1, F, G, H, I
Implementation-neutral throughout	PASS — no programming language, framework, or database product named

26. Assumptions

- The Controller-Service-Repository pattern (Section 6) is a conceptual organizing structure only; the eventual Physical Design/implementation may use a different concrete pattern (e.g., CQRS, hexagonal architecture) as long as the same separation-of-concerns principle (Section 3) is preserved.
- All 71 Business Rules from Appendix-C v1.1 are assumed enforceable at the Service layer without requiring database-vendor-specific stored-procedure logic, consistent with the "no stored procedures" scope exclusion (Section 2.2).
- Event Publisher/Consumer (Section 5) is assumed implementable via any suitable asynchronous mechanism (in-process event bus, message queue, etc.); the specific mechanism is a Physical Design decision, not fixed here.
- The single-institute working assumption from Appendix-I Section 21.2 continues to apply; no module's internal design introduces institute-scoping logic at this stage.
- All numeric/policy values already marked Client/Product Decision Required across Appendix-C v1.1, Appendix-F, Appendix-G, Appendix-H v1.1, and Appendix-I remain open at this LLD stage and are not resolved here — resolving them is a client/product decision, not a design-team decision.

27. Final Review

Metric	Value
Modules Detailed (Section 4)	11 of 11 requested
Conceptual Component Types (Section 5)	6 (Controller, Service, Validator, Repository, Event Publisher/Consumer, Background Job)
Background Jobs Identified (Section 17)	8
Required ASCII Diagrams	4 of 4 (Component, Module Interaction, Internal Processing Flow, Layer Interaction) — present in Sections 6, 7, 18
Traceability to Appendix-I	Complete — Section 24
No Fabricated Scope	Confirmed — every module, entity, rule, role, and integration reused from Appendix-E/F/G/H/I by ID
LLD Quality Score	90/100
Approval Status	Conditionally Ready — internal design is structurally complete for all 11 modules; approval is gated on the same consolidated Client/Product Decision Required items already tracked across Appendix-C v1.1, F, G, H, and I, since several (statutory slabs, thresholds, vendor selections) directly affect Service-layer implementation detail

Deductions from a perfect score reflect the volume of downstream Client/Product Decision Required items inherited from earlier appendices, not any structural incompleteness in this LLD's own 27 sections.